Construct the network and calculate the earliest start time, earliest finish time, latest start time and latest finish time of each activity and determine the Critical path of the project and duration to complete the project.

10. The following table use the activities in a construction projects and relevant information

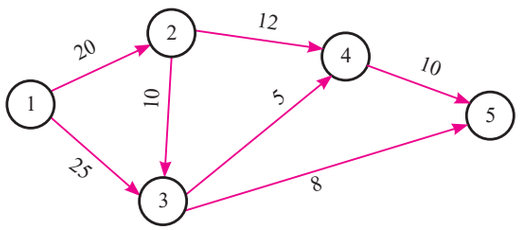
Activity	1-2	1-3	2-3	2-4	3-4	4-5
Duration (in days)	22	27	12	14	6	12

Draw the network for the project ,calculate the earliest start time, earliest finish time, latest start time and latest finish time of each activity and find the critical path. Compute the project duration.

Exercise 10.3

Choose the correct answer

1. The critical path of the following network is



(a) 1 – 2 – 4 – 5 (b) 1– 3– 5
(c) 1 – 2 – 3 – 5 (d) 1 – 2 – 3 – 4 – 5

2. Maximize: $z = 3x_1 + 4x_2$ subject to $2x_1 + x_2 \leq 40$, $2x_1 + 5x_2 \leq 180$, $x_1, x_2 \geq 0$. In the LPP, which one of the following is feasible corner point?
(a) $x_1 = 18, x_2 = 24$
(b) $x_1 = 15, x_2 = 30$
(c) $x_1 = 2 \cdot 5, x_2 = 35$
(d) $x_1 = 20 \cdot 5, x_2 = 19$

3. One of the conditions for the activity (i, j) to lie on the critical path is
(a) $E_j - E_i = L_j - L_i = t_{ij}$
(b) $E_i - E_j = L_j - L_i = t_{ij}$
(c) $E_j - E_i = L_i - L_j = t_{ij}$
(d) $E_j - E_i = L_j - L_i \neq t_{ij}$

4. In constructing the network which one of the following statement is false?
(a) Each activity is represented by one and only one arrow. (i.e.) only one activity can connect any two nodes.
(b) Two activities can be identified by the same head and tail events.
(c) Nodes are numbered to identify an activity uniquely. Tail node (starting point) should be lower than the head node (end point) of an activity.
(d) Arrows should not cross each other.

5. In a network while numbering the events which one of the following statement is false?
(a) Event numbers should be unique.
(b) Event numbering should be carried out on a sequential basis from left to right.
(c) The initial event is numbered 0 or 1.
(d) The head of an arrow should always bear a number lesser than the one assigned at the tail of the arrow.

6. A solution which maximizes or minimizes the given LPP is called

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